STUDENT PROJECT 3

Title

Design, Fabrication, and Flight Testing of a Foamboard RC Trainer Aircraft

Project Type

Design–Build–Test (DBT) Based Mini Project

Objective

The objective of this project is to provide students with hands-on experience in **basic aircraft design, lightweight structure fabrication, propulsion system selection, and flight control integration** by designing and building a **4-channel foamboard radio-controlled (RC) trainer aircraft**. The project emphasizes stability-oriented design, safe operation, and experimental validation through flight testing.

Project Description

In this project, students will design and fabricate a **high-wing, fixed-wing RC aircraft** using low-cost foamboard materials. The aircraft will be powered by an electric propulsion system and controlled using **ailerons, elevator, rudder, and throttle**. Students will analyze the aircraft geometry, estimate weight and balance, select suitable electronics, fabricate the airframe, and evaluate flight performance through controlled test flights.

The project is intended to introduce fundamental concepts of **aerodynamics, flight stability, structural design, and control systems**, while promoting teamwork and practical problem-solving skills.

Design Specifications

Airframe

1. Wingspan: **1050 mm**
2. Fuselage length: **720 mm**
3. Construction material: **5 mm foam board (or XPS foam)**
4. Configuration: **High-wing monoplane**
5. Landing gear: **Tricycle configuration**
6. Foam requirement:
    a. 2 sheets of A1 size **or**
    b. 3 sheets of A2 size

Comprehensive details of the RC aircraft design and fabrication plan are presented in the Appendix for reference

Weight Constraints

1. Maximum airframe weight (without battery): **560 g**
2. Maximum take-off weight (with battery): **860 g**

Control System

1. Number of channels: **4**
    a. Ailerons
    b. Elevator
    c. Rudder
    d. Throttle
2. Control surface deflections:
    a. Ailerons: ±17 mm
    b. Elevator: ±20 mm
    c. Rudder: ±25 mm
3. Radio exponential:
    - Ailerons: **70%**

Propulsion System

1. Motor: **2206 brushless DC motor (1600–2600 KV)**
2. Propeller: **5″–7″ (as recommended by motor manufacturer)**

Battery Options

1. **Beginner configuration:**
    a. 2S Li-Po, 2200–4000 mAh
    b. Calm weather operation only
2. **High-power configuration:**
    a. 3S Li-Po, 1500–3200 mAh
    b. Suitable for light wind and maneuvering

Construction Guidelines

- Recommended adhesive: **Hot glue**
- Avoid aggressive adhesives such as cyanoacrylate (CA) glue
- Ensure structural reinforcement at wing spars, landing gear mounts, and motor firewall

Project Tasks

Students are required to complete the following stages:

Phase 1 – Conceptual Design

1. Study trainer aircraft configurations
2. Interpret given three-view aircraft drawings
3. Identify major aerodynamic and structural components

Phase 2 – Preliminary Analysis

1. Estimate:
    a. Wing loading
    b. Thrust-to-weight ratio
    c. Center of gravity (CG) location
2. Justify motor, propeller, and battery selection

Phase 3 – Fabrication

1. Cut and assemble foamboard airframe
2. Install control surfaces and hinges
3. Mount propulsion and landing gear

Phase 4 – Electronics Integration

1. Install motor, ESC, receiver, servos, and battery
2. Configure transmitter control rates and exponential
3. Perform ground testing and control surface checks

Phase 5 – Flight Testing and Evaluation

1. Conduct taxi tests and short hops
2. Perform steady-level flight tests
3. Evaluate stability, control response, and flight duration

Deliverables

Each student group must submit:

1. **Project Report** (hard copy / PDF) containing:
    a. Introduction and objectives
    b. Design methodology
    c. Calculations and assumptions
    d. Fabrication process
    e. Flight test observations
    f. Conclusions and learning outcomes
2. **Aircraft Model**
    a. Fully assembled and airworthy RC aircraft
3. **Flight Test Evidence**
    - Photographs or videos of flight tests

Evaluation Criteria

Component	**Weightage**
Design & Analysis	30%
Fabrication Quality	30%
Flight Performance	20%
Report Quality	20%

Learning Outcomes

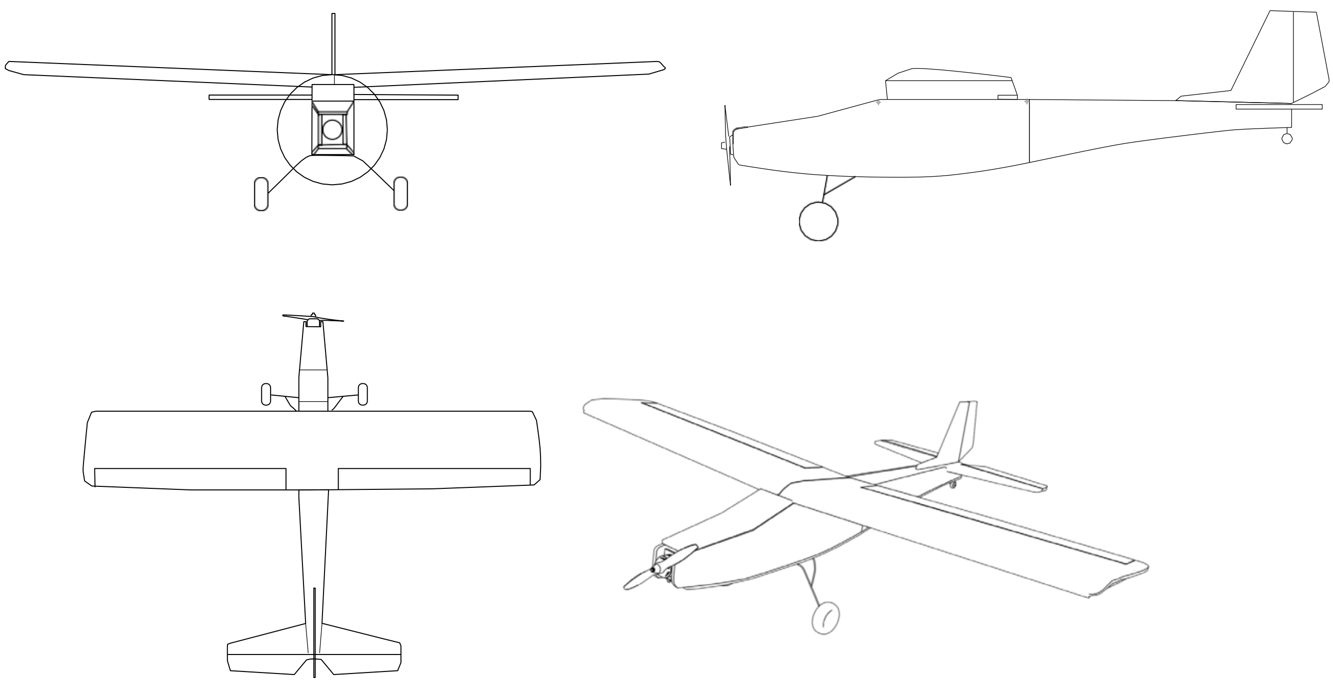
Upon completion of this project, students will be able to:

1. Understand basic aircraft stability and control concepts
2. Apply lightweight structural design techniques
3. Select appropriate propulsion and control systems
4. Integrate electronics into an air vehicle
5. Perform experimental flight testing and analysis

Safety Instructions

1. All flights must be conducted in an open field
2. Battery handling and charging must follow Li-Po safety guidelines
3. Instructor approval is mandatory before maiden flight

Appendix
Mini project-3



The final 3D views of the model after preparation from the plan given in the appendix.

Take print out of the plan in A4 sheets and join them to form the plans for various parts of the Airplane.

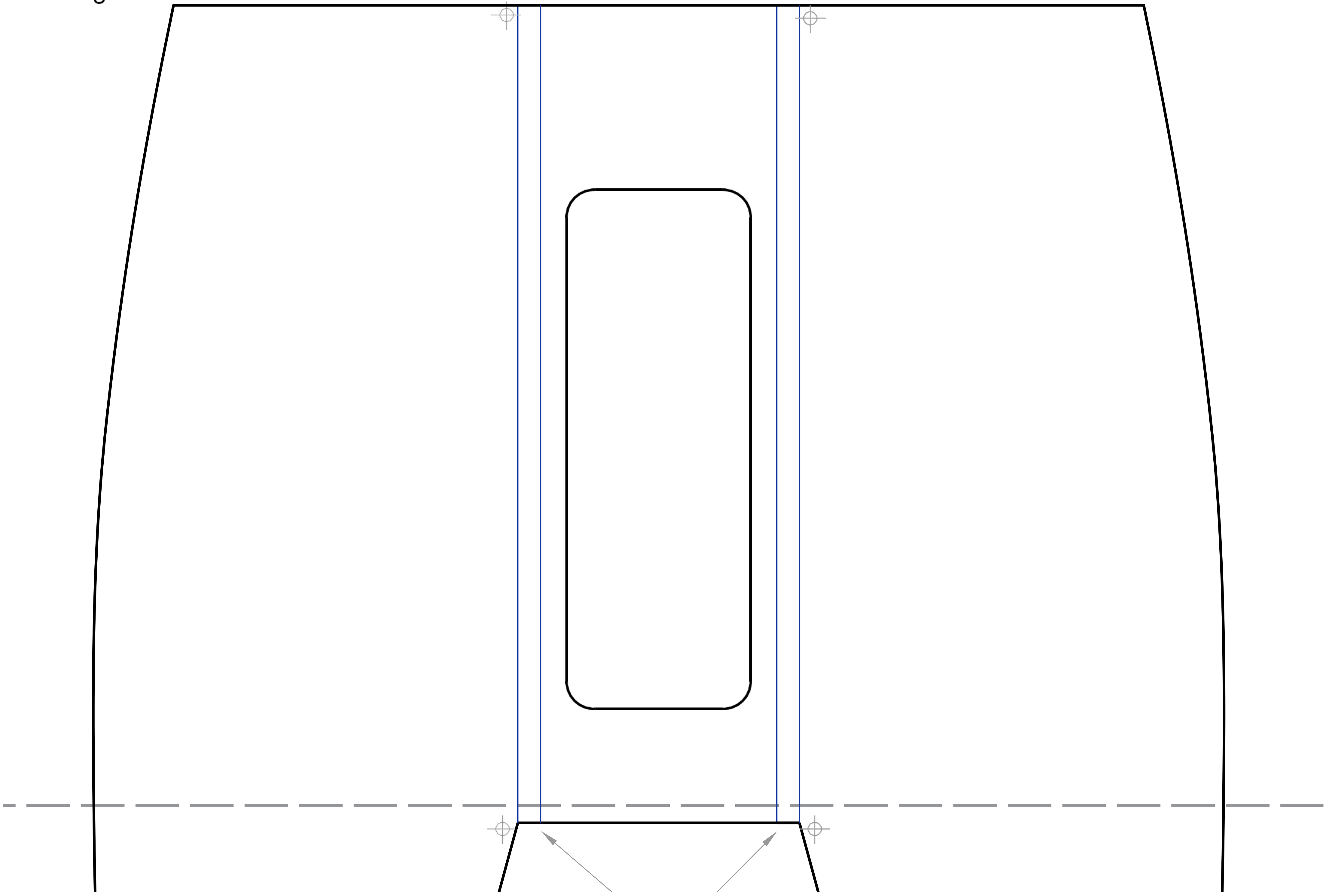
Dark lines Full cut

Blue lines 50% cut with 450 chamfer for bending and forming the shape

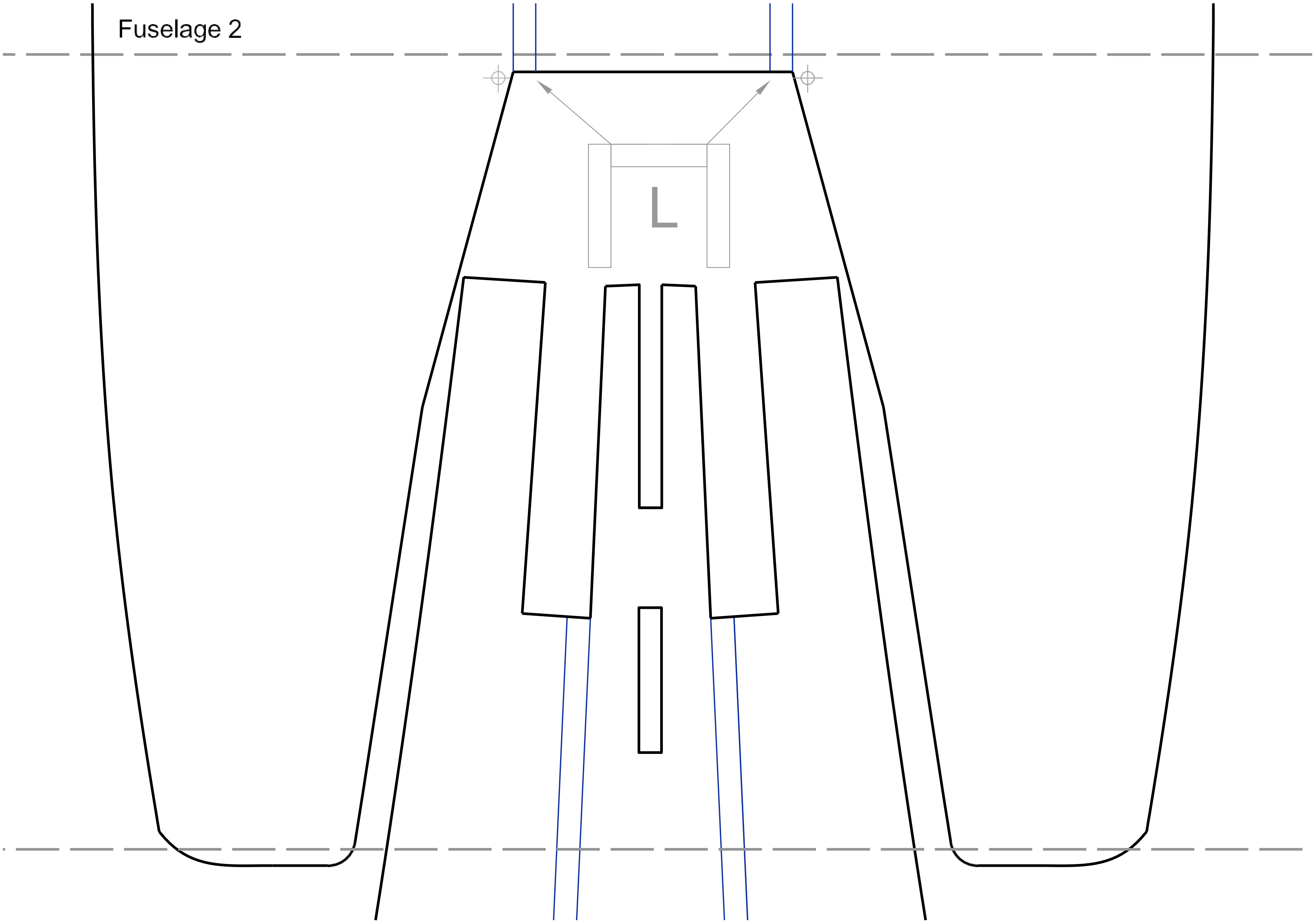
Dotted line are used for joining.

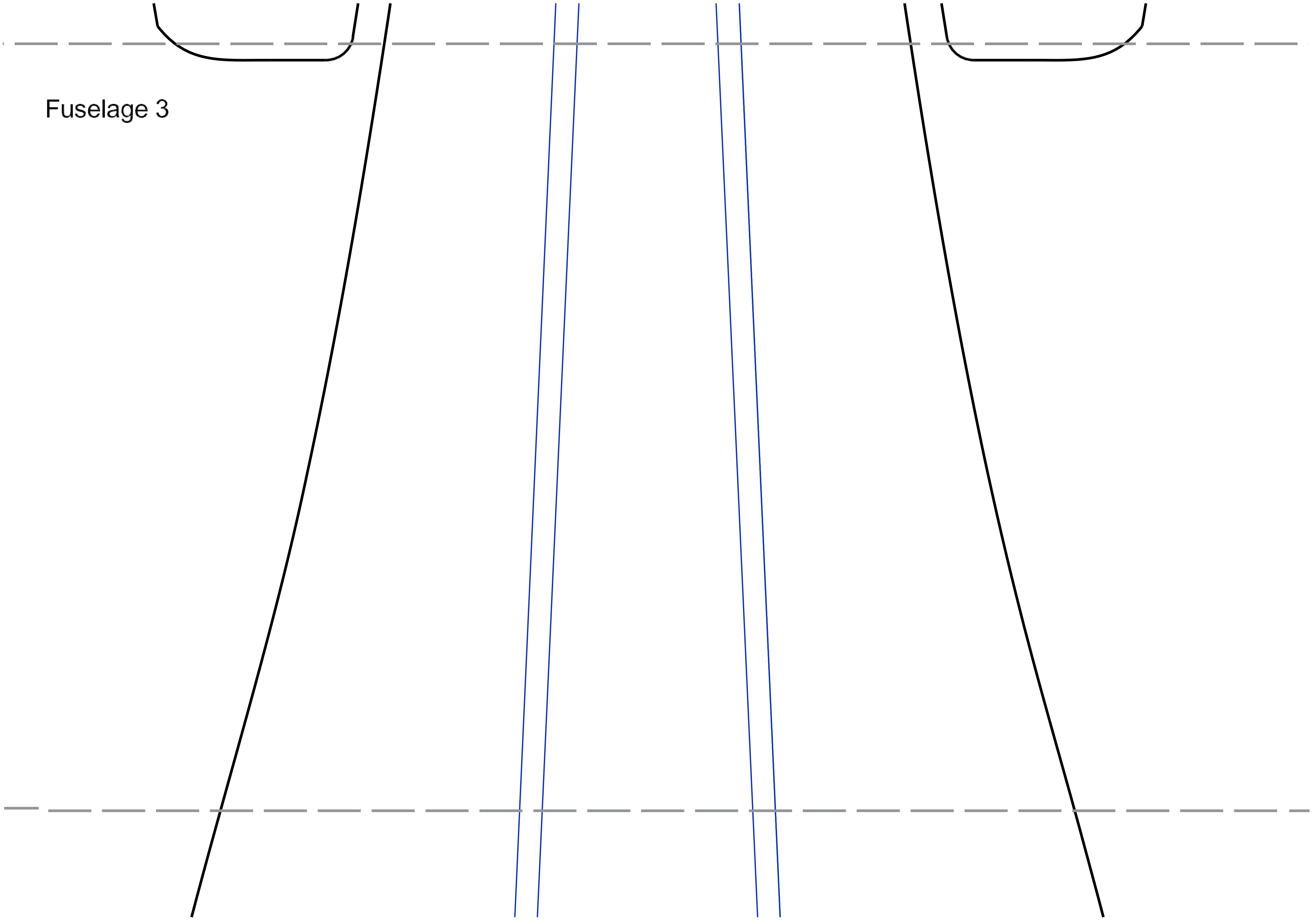
For wing only one plan is given take extra print for the wing .

Fuselage 1

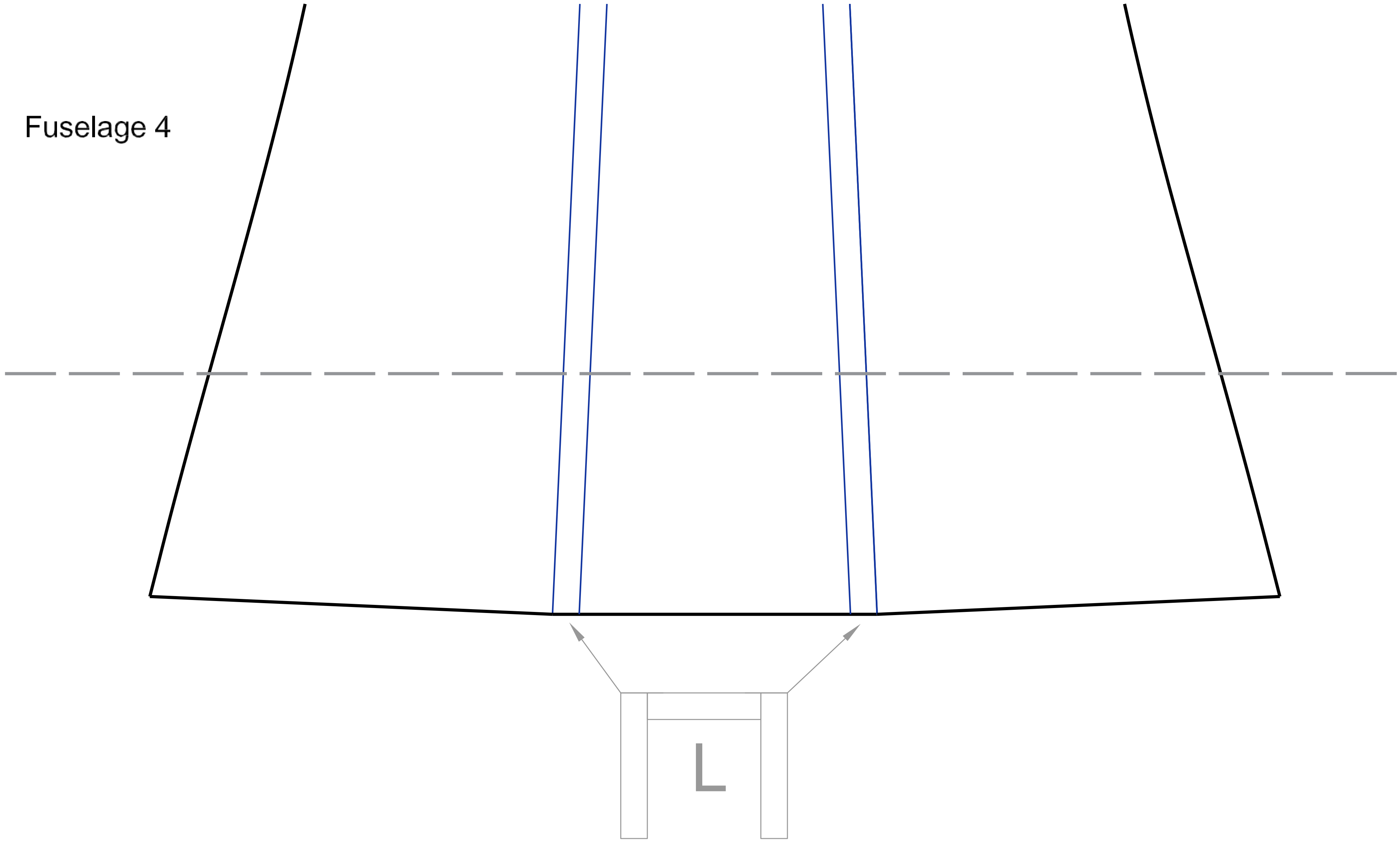


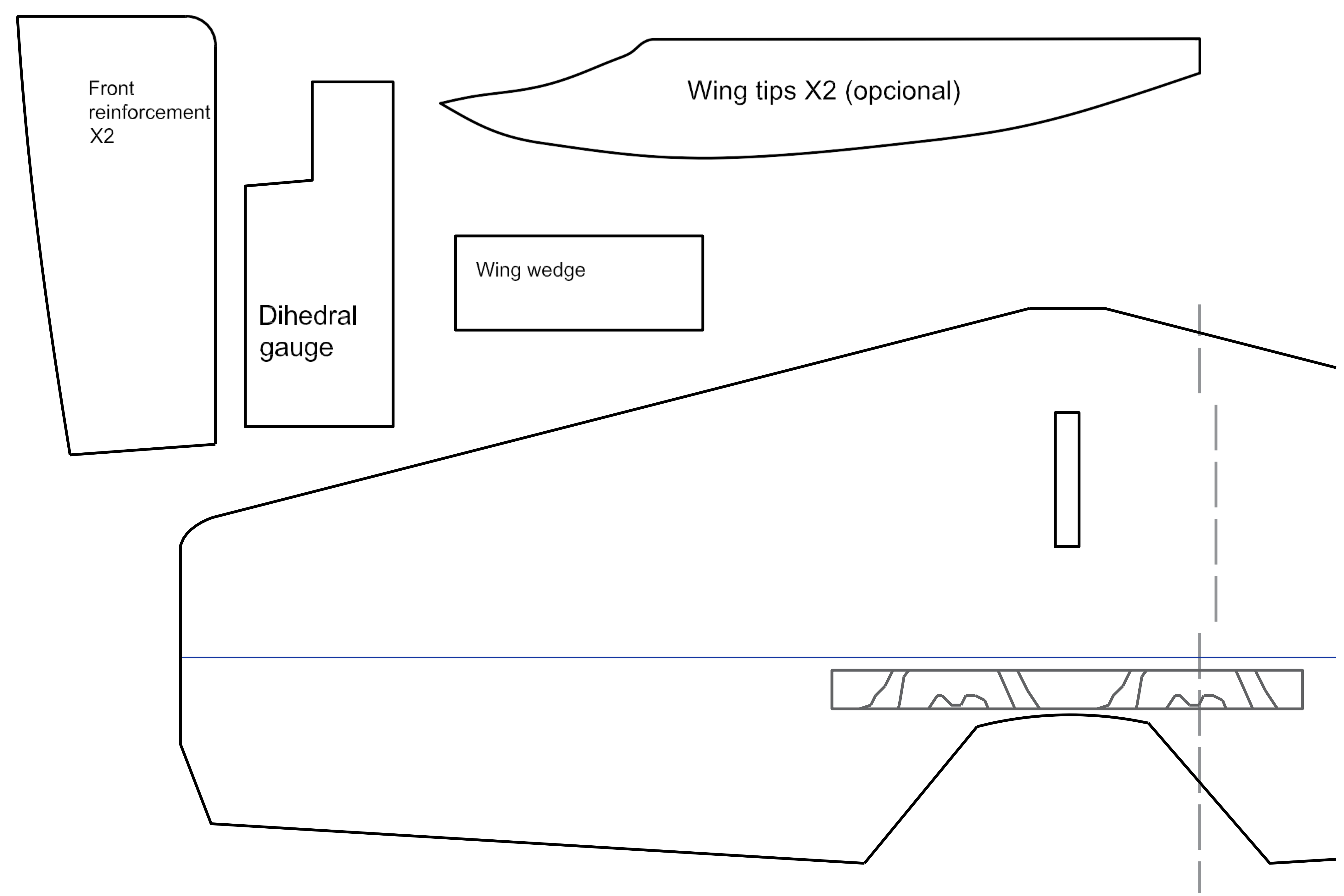
Fuselage 2



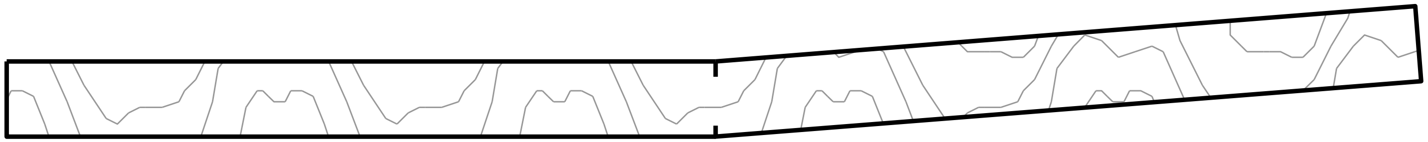

Fuselage 3

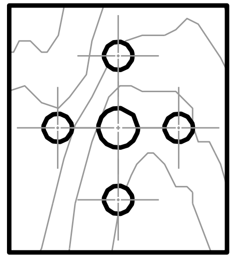
Fuselage 4



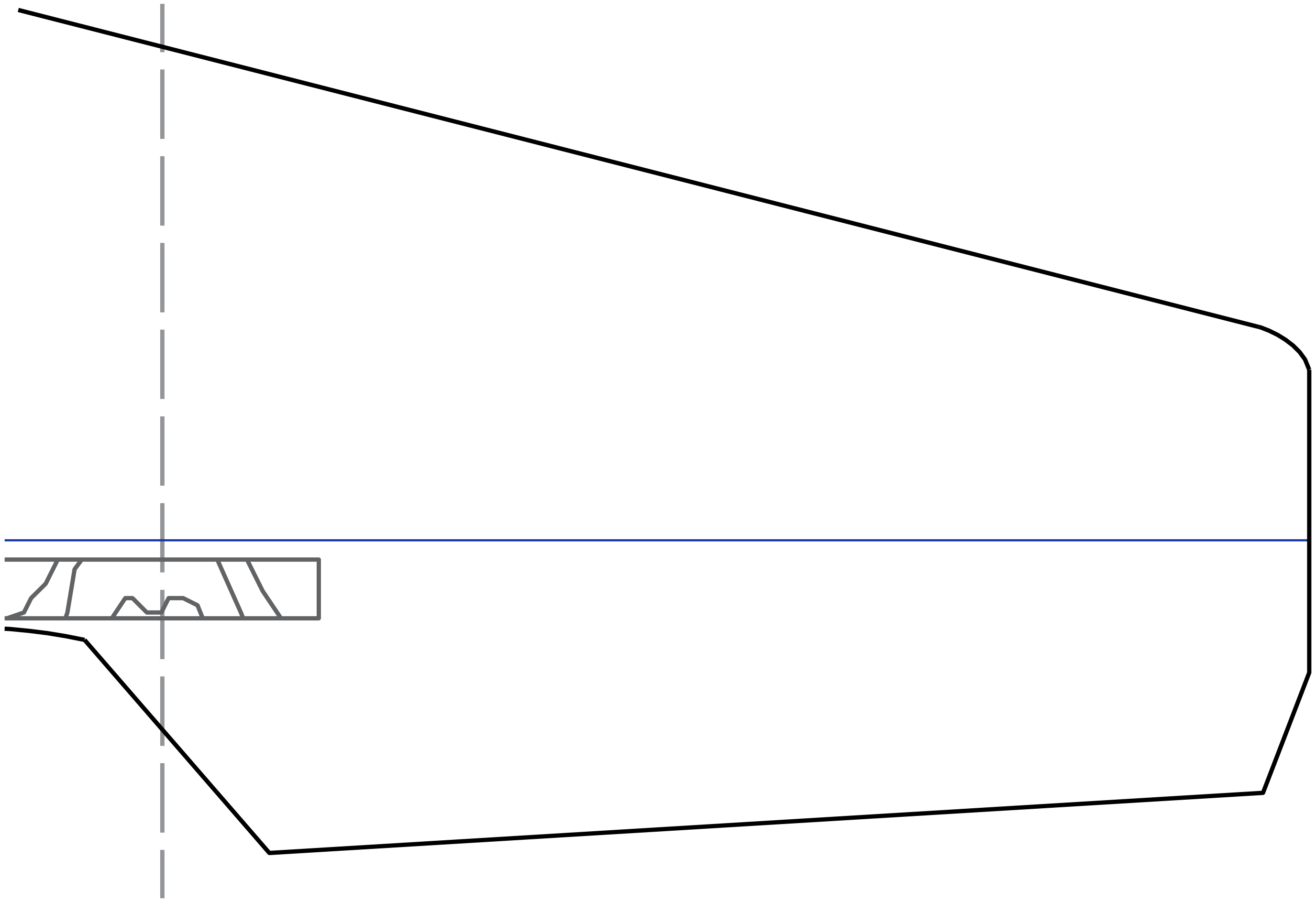

Front
reinforcement
X2
Dihedral
gauge
Wing tips X2 (opcional)
Wing wedge

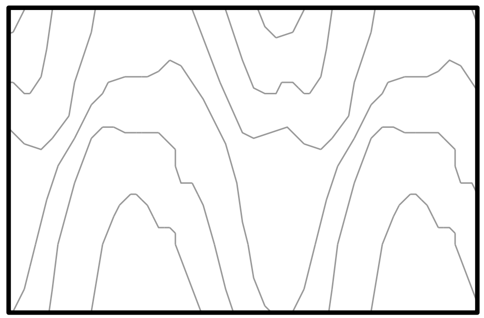
Central spar (plywood) 3-5mm



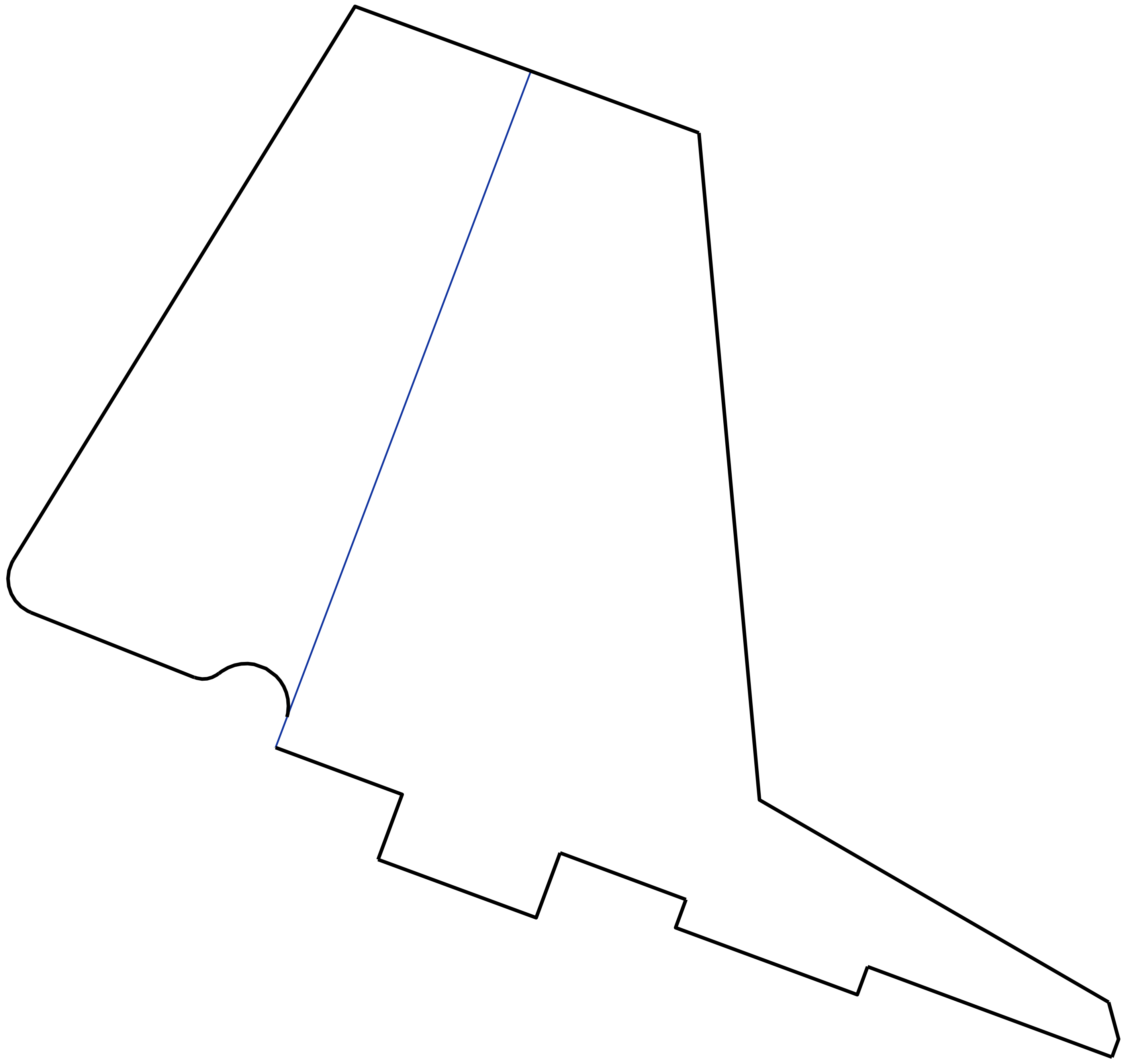


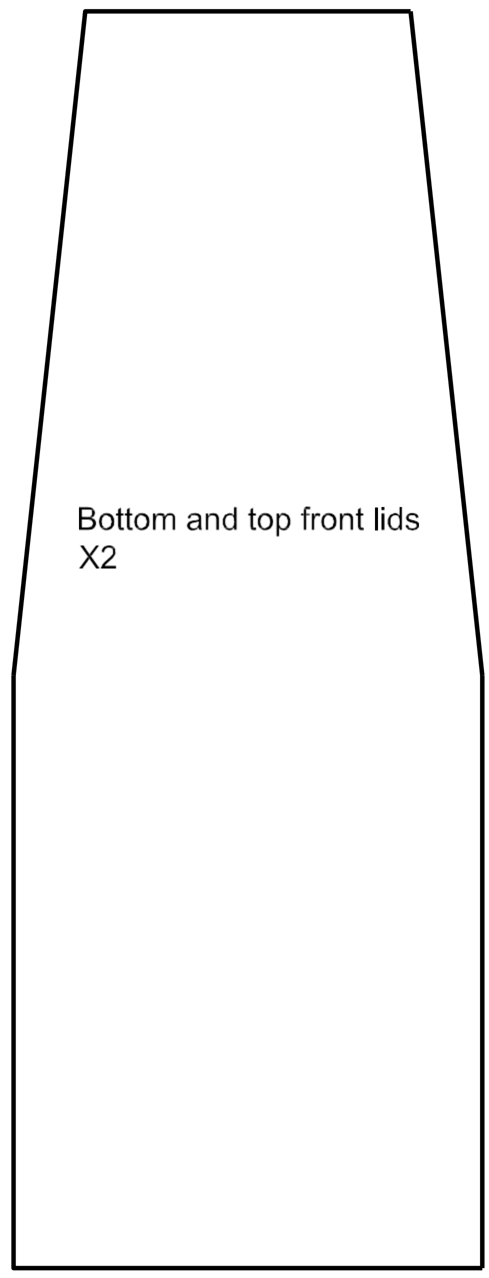
Motor base (Plywood 3-5mm)

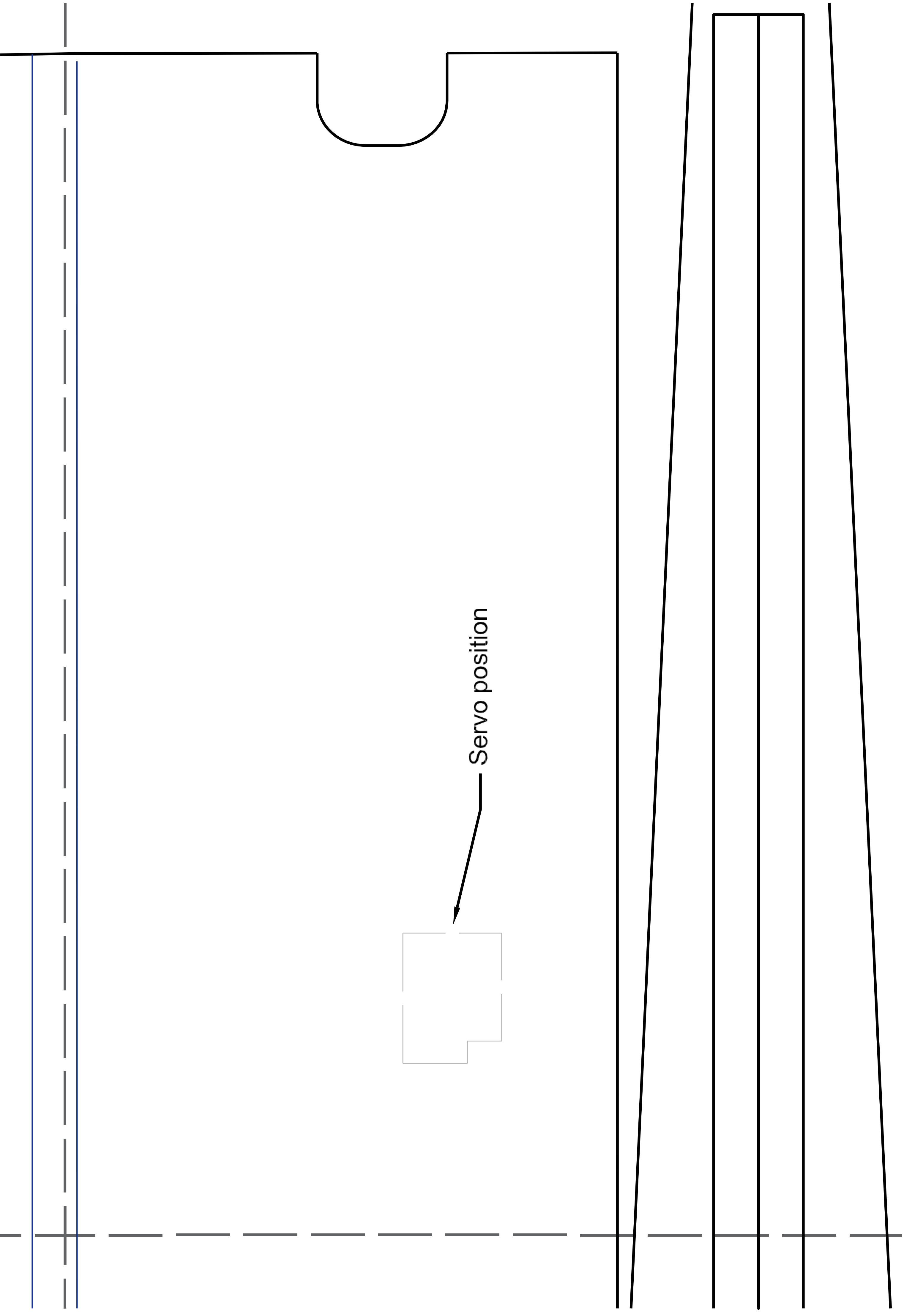


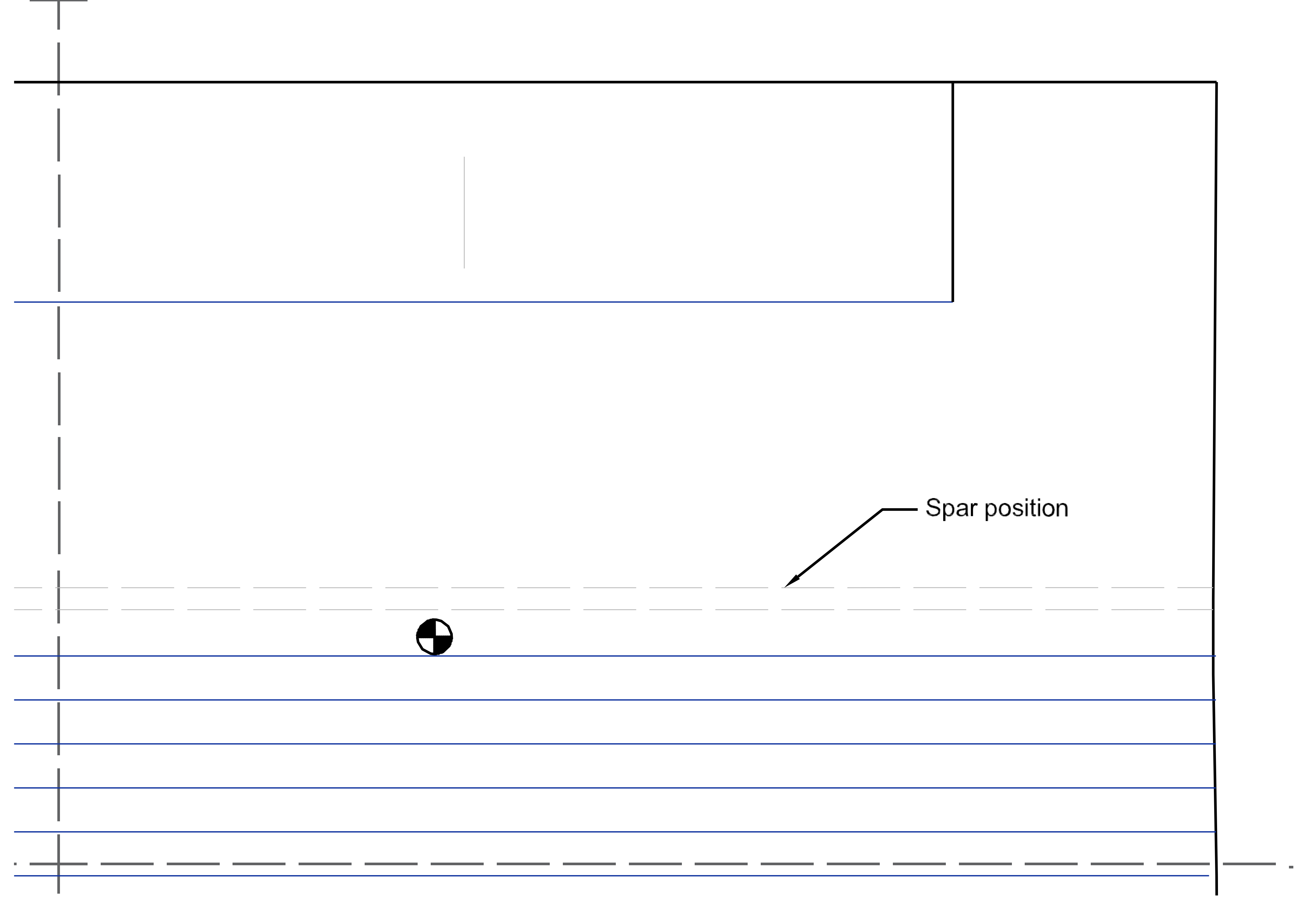


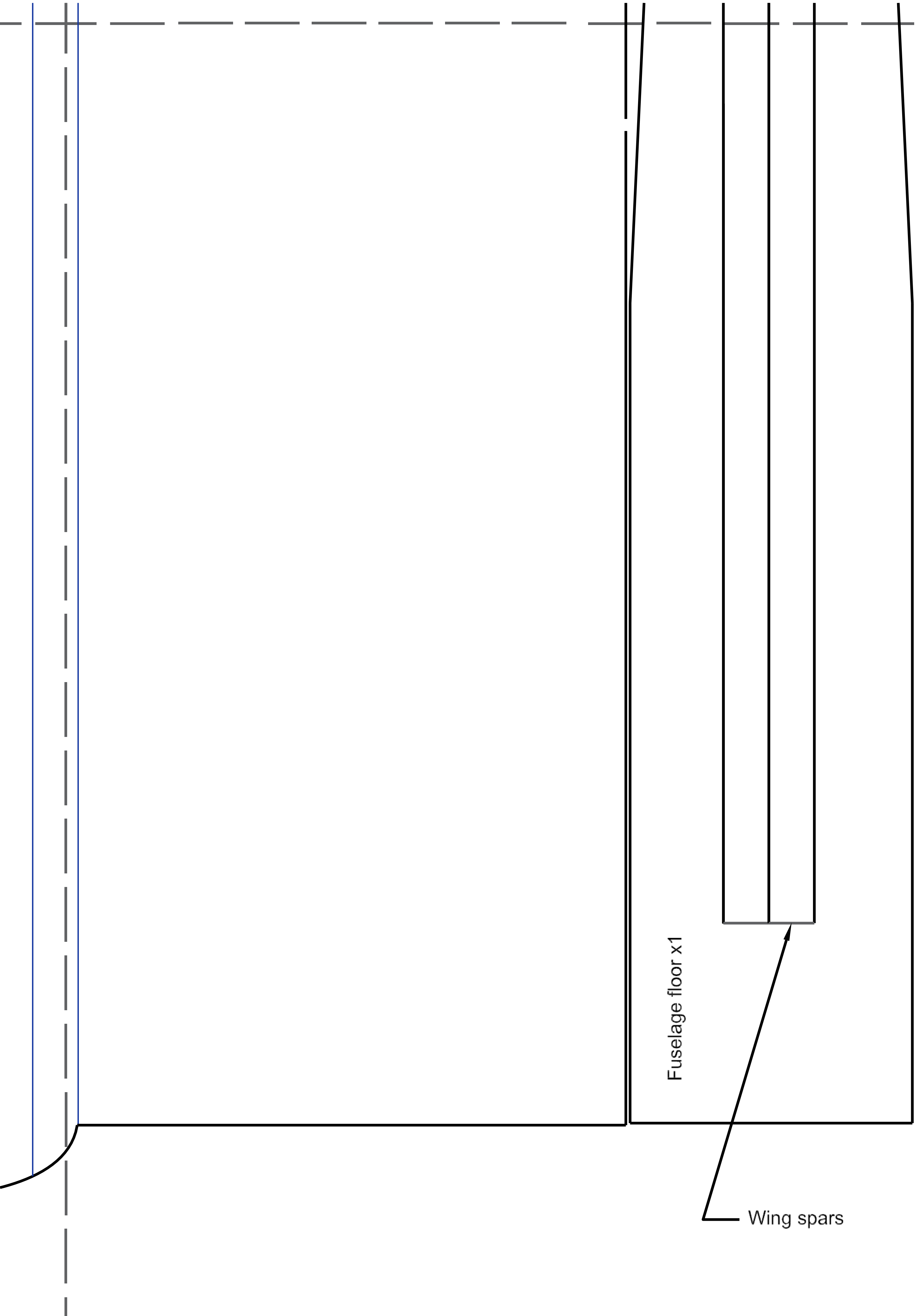
Landing gear base (plywood 3-5mm)




Bottom and top front lids
X2


Servo position


Spar position


Fuselage floor x1
Wing spars